WATER SUPPLY COMPLAINT & BILLING SYSTEM

SDLC Project Scheduling — Based on System Request & Feasibility Study

System Activities Table

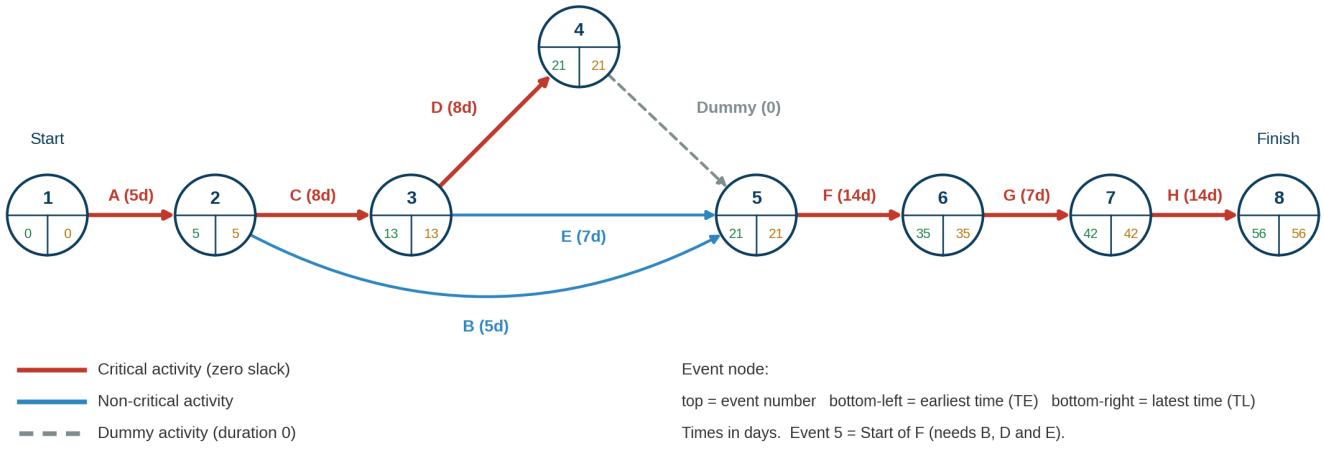
The following table maps system development tasks based on the Water Supply Complaint & Billing System analysis:

Activity	Description (Derived from System Request & Feasibility)	Predecessor	Duration (Days)
A	System Request & Scope Definition (business need, requirements, constraints)	-	5
B	Feasibility Study (Technical, Economic & Organizational)	A	5
C	Requirements Elicitation (Complaint Workflow, Meter Reading & Billing Rules)	A	8
D	Database Schema Design (Customers, Meters, Readings, Bills, Complaints)	C	8
E	UI/UX Design (Customer Portal / Mobile App & Management Dashboard)	C	7
F	Core Development: Complaint Engine (Tracking No., Auto-Routing, Duplicate Merging), Billing Module, Portal / App & Management Reports	B, D, E	14
G	Integration & Security: SMS / E-bill, Online Payment Gateway, Existing Customer & Meter Data, Data Protection	F	7
H	System Testing, UAT, Staff Training & Pilot Rollout (one branch first)	G	14

Activity H needs F and G; since G already depends on F, the predecessor is written as G only (F is implied).

TASK 1 & TASK 2: PERT DIAGRAM SIMULATION

Draw the PERT network diagram representing activities sequence and dependencies:



Activities D and E both start after C and both must finish before F, so one **dummy activity** (duration 0) is added from event 4 to event 5 to keep every activity uniquely identified by its start and end events.

TASK 3: GANTT CHART & MINIMUM PROJECT DURATION

Requirements: Draw the Gantt Chart, calculate the Critical Path, and determine the Minimum Total Execution Time for the Water Supply Complaint & Billing System.

Critical Path & Duration Analysis

Possible Paths Comparison:

Path 1: A → B → F → G → H = 5 + 5 + 14 + 7 + 14 = 45 Days

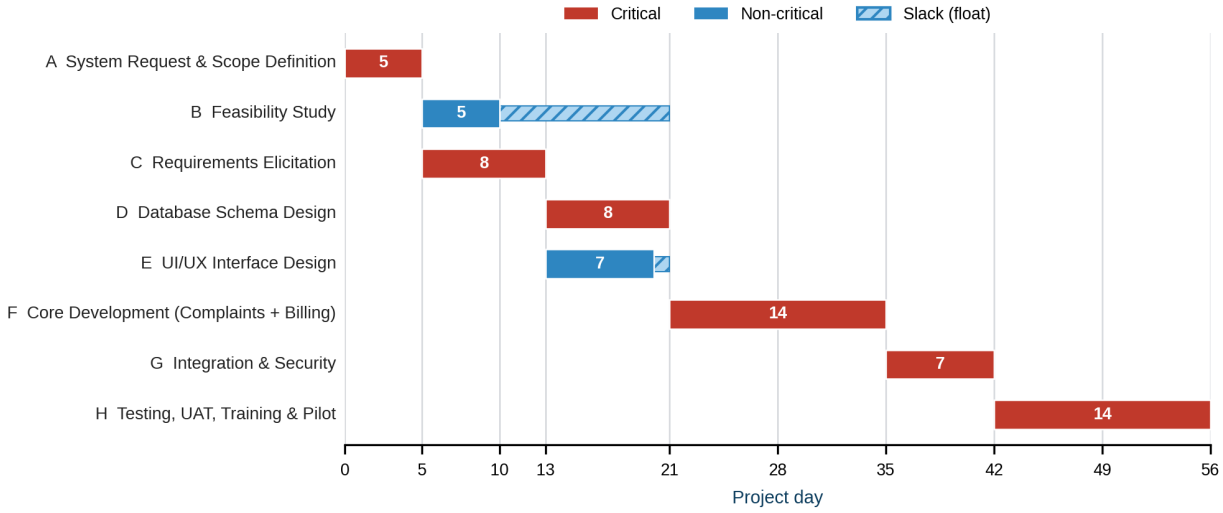
Path 2: A → C → D → F → G → H = 5 + 8 + 8 + 14 + 7 + 14 = **56 Days (Longest Path)**

Path 3: A → C → E → F → G → H = 5 + 8 + 7 + 14 + 7 + 14 = 55 Days

Critical Path: A → C → D → F → G → H

Minimum Project Completion Time: 56 Days (8 weeks)

Activity E has 1 day of slack and activity B has 11 days of slack; delaying any activity on the critical path delays the whole project.



Schedule Table (forward and backward pass)

Activity	Earliest Start	Earliest Finish	Latest Start	Latest Finish	Slack	Critical?
A	0	5	0	5	0	Yes
B	5	10	16	21	11	No
C	5	13	5	13	0	Yes
D	13	21	13	21	0	Yes
E	13	20	14	21	1	No
F	21	35	21	35	0	Yes
G	35	42	35	42	0	Yes
H	42	56	42	56	0	Yes

Earliest/Latest times are measured in project days from Day 0. Slack = Latest Start − Earliest Start.